Student Answer Script Transcription (High-Resolution Stitched Document)

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(v) Use extra pages, if necessary, towards the booklet.
Q-1) Space for answering Question-1 only.
a) For S to be subspace of $M_{4\times4}(R)$, it must satisfy the following:
i) S must include zero matrix.
In zero matrix, as all the elements are zero,
the condition for S is satisfied $(0+0+0=0)$ for fourth column.

Question 1

Let W_1 and W_2 be two elements from S. Then $W_1 + W_2$ must belong to S.

Each element in $W_1 + W_2$ is the sum of respective elements of W_1 and W_2. So, the fourth column of $(W_1 + W_2)$ = Fourth col. of W_1 + Fourth col. of W_2, which can be further broken down

Question:

Which can be further broken down to sum of first 3 columns of W_1 + sum of first 3 columns of W_2? By associativity of addition, $(W_1 + W_2)$ belongs to S. If $W \in S$, then tW must belong to S^T. As we in row $\begin{bmatrix} a_1, a_2, a_3 \\ a_4, a_5, a_6 \end{bmatrix}$, $ta_1 + ta_2 + ta_3 = t(a_1 + a_2 + a_3)$ (commutative).

Answer:

The student's answer is as follows:

(i)

The student wrote: **Sum of First 3 Columns of W_1 + Sum of First 3 Columns of W_2**

(ii)

The student wrote: **By associativity of addition,** $(W_1 + W_2)$ **belongs to** S**. If** $W \in S$**, then** tW **must belong to** S^T**. As we in row** $\begin{bmatrix} a_1, a_2, a_3 \\ a_4, a_5, a_6 \end{bmatrix}$**,** $ta_1 + ta_2 + ta_3 = t(a_1 + a_2 + a_3)$ **(commutative).**

Conclusion:

The student correctly identified the sum of the first three columns of W_1 and W_2 and applied the properties of addition and commutativity to show that tW must belong to S^T. The answer is well-structured and demonstrates understanding of the concepts involved.